

Type: Practical(Minor) B.Sc(ECS)-II (Semester IV) Course Title: Practical based on DSC2-5		(Paper Code:)
Total Lectures: 30 Hrs.		Contact Hrs. (P) : 2
University Evaluation: 15 Marks		Internal Evaluation: 10 Marks
Tools: Python / Matlab		
1.	Implement a Python function to handle similar problems using factorials.	
2.	Generate all possible arrangements of the word “ DATA ”.	
3.	Calculate the number of 4-digit PIN codes that can be formed using the digits 0-9 (repetition allowed).	
4.	Generate all possible 3-letter passwords from the alphabet (without repetition).	
5.	Simulate 1,000 coin flips and calculate the probability of heads and tails.	
6.	Simulate rolling a die 1,000 times and calculate the probability of each face (1-6).	
7.	Simulate binomial trials (e.g., number of successes in 10 coin flips over 1,000 trials).	
8.	Model the number of emails received per hour using a Poisson distribution.	
9.	Conduct a one-sample t-test to determine if the average test score of a class differs from a national average.	
10.	Conduct a chi-square test for independence using a categorical dataset.	